Squeezing for Broadband Multidimensional Variational Measurement

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Broadband multidimensional variational measurement allows to overcome Standard Quantum Limit (SQL) of a classical mechanical force detection, resulting from quantum back action, which perturbs evolution of a mechanical oscillator. In this optomechanic scheme detection of a resonant signal force acting on a linear mechanical oscillator coupled to a system with three optical modes with separation nearly equal to the mechanical frequency. The measurement is performed by optical pumping of the central optical mode and measuring the light escaping the two other modes. Detection of optimal quadrature components of the optical modes and post processing result in the back action exclusion in a broad frequency band and surpassing SQL. We show that optical losses inside cavity restrict back action exclusion due to loss noise. We also analyze how two-photon (nondegenerate) and conventional (degenerate) squeezing improve sensitivity with account optical losses, considering mainly internal squeezing.

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I. INTRODUCTION

Optical transducers are frequently used to observe mechanical motion. These transducers allow to detect displacement, speed, acceleration, and rotation of mechanical systems. Mechanical motion can change phase or amplitude of the probe light. The sensitivity of the measurement can be extremely high, for example, a relative mechanical displacement much smaller than a proton size can be detected. It was demonstrated in gravitational wave detectors [1–7], in magnetometers [8, 9], and in torque sensors [10–12].

There are several reasons restricting the fundamental sensitivity of the measurement. One of them is the fundamental thermal fluctuations in mechanical system (Nyquist noise). However, this obstacle can be considerably decreased if one measures a variation of the position during time much smaller than the system ring down time [13, 14].

Another limitation comes from the quantum noise of the meter. On one hand, the accuracy of the measurements is limited because of their fundamental quantum fluctuations, represented by the shot noise for the optical probe wave, in order to decrease it one have to increase pump power. On the other hand, the sensitivity is impacted by the perturbation of the state of the probe mass due to so called "back action", which increases with growth of pump power. In the case of optical meter the mechanical perturbation results from fluctuations of the light pressure force. Interplay between these two factors leads to a so called standard quantum limit (SQL) [13, 14] of the sensitivity.

SQL is a consequence of noncommutativity between the probe noise and the quantum back action noise. In a simple displacement sensor the probe noise is the phase noise of light and the back action noise is the amplitude noise of light (light pressure noise). The signal is contained in the phase of the probe. The relative phase noise decreases with optical power whereas the relative back action noise increases with the power. The optimal measurement sensitivity corresponds to SQL. It is not possible to measure the amplitude noise and subtract it from the measurement result, because of phase and amplitude quantum fluctuations of the same wave do not commute.

The SQL of a mechanical force, acting on free test mass, can be surpassed in a transducer supporting opto-mechanical velocity measurement [15, 16]. The limit also can be overcome using opto-mechanical rigidity [17, 18]. Preparation of the probe light in a nonclassical squeezed state [19–26] as well as detection of a variation of a strongly perturbed optical quadrature [27–30] curbs the quantum back action and lifts SQL. The SQL can be surpassed with coherent quantum noise cancellation [31–33] as well as compensation using an auxiliary medium with negative nonlinearity [34]. Optimization of the measurement scheme by usage a few optical frequency harmonics as a probe also allows beating the SQL. A dichromatic optical probe may lead to observation of such phenomena as negative radiation pressure [35, 36] and optical quadrature-dependent quantum back action evasion [37].

The first procedure of back action evading (BAE) for mechanical oscillator, proposed about forty years ago [38, 39], took advantage of short (stroboscopic) measurements of a mechanical coordinate separated by a half period of oscillator. At the same time it was proposed to measure not a coordinate but one of quadrature amplitudes of a mechanical oscillator [38, 40] to perform a BAE. Both propositions are equivalent and can be realized with a pulsing pump [39, 41, 42].

Recently broadband multidimensional variational measurement for mechanical oscillator was proposed [43]. Mechanical oscillator is coupled to a system with three optical modes, which frequencies $\omega_\pm$, ω_0 are separated by the mechanical frequency ω_m: $\omega_\pm = \omega_0 \pm \omega_m$. The measurement is performed by optical pumping of the central optical mode ω_0 and measuring the light escaping the two other modes $\omega_\pm$. Detection of optimal quadrature components of the output waves of modes $\omega_\pm$ *separately* provides two channel registration. It allows to detect back

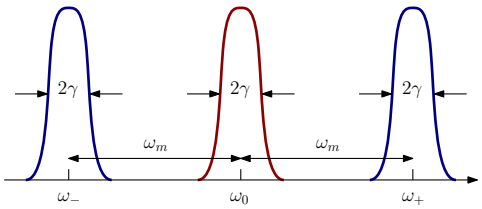

Figure 1. Optomechanical scheme. Three optical modes which frequencies are separated by frequency ω_m of mechanical oscillator. Optical modes are coupled with the mechanical oscillator. Relaxation rate γ is the same for the all three modes, $\gamma \ll \omega_m$. The middle mode with frequency ω_0 is resonantly pumped.

action and to remove it completely from the measured data via post processing.

In quantum measurement of free mass position preparation of the probe light in a squeezed state [19–25, 44–46] gives possibility to surpass SQL. In this paper we analyze how sensitivity of broadband multidimensional variational measurement for detection of small signal force, acting on mechanical oscillator, can be improved by squeezing even in presence of optical losses. We consider two possibilities of two photon (nondegenerate) and conventional (degenerate) *internal* squeezing and compare them with each other. We also analyze and compare sensitivity improvement for internal squeezing for the cases of two photon and conventional squeezing.

II. PHYSICAL MODEL

Let we have cavity with triplet of optical modes ω_-, ω_0, ω_+ separated from each others by eigen frequency ω_m of mechanical oscillator as shown on Fig. 1. The middle mode with frequency ω_0 is resonantly pumped, the modes $\omega_\pm$ are not pumped. Mass of mechanical oscillator is a movable end mirror, it provides coupling with optical modes. We assume two photon (nondegenerate) *internal* squeezing of fields in the modes $\omega_\pm$, created by parametric interaction with pump on frequency $2\omega_0$. We detect output of modes $\omega_\pm$. See scheme on Fig. 2.

We assume that the relaxation rates of the optical modes are identical and characterized with the full width at the half maximum (FWHM) equal to $2\gamma = 2(\gamma_0 + \gamma_e)$, where γ_0 characterizes transmittance of input mirror and γ_e — optical losses of cavity's mirrors, relaxation rates γ_0, γ_e are the same for all optical modes. The mechanical relaxation rate γ_m is small as compared with the optical one. We also assume that frequency synchronization, the conditions of the resolved side band interaction and condition of optical loss smallness are valid:

$$\omega_\pm = \omega_0 \pm \omega_m, \quad \gamma_m \ll \gamma \ll \omega_m, \quad \gamma_e \ll \gamma_0 \tag{2.1}$$

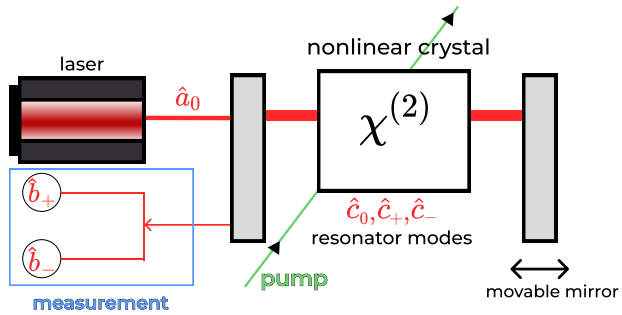

Figure 2. An optomechanical scheme based on a Fabry-Perot resonator with a transmissive front mirror and a non-transmissive movable rear wall. There is a nonlinear crystal inside the resonator, which is pumped at twice the laser carrier frequency to create two photon squeezing in modes $\omega_\pm$ of the resonator. The output of these modes are then measured separately, details of the measurement are shown in Fig. 3.

A. Hamiltonian

The generalized Hamiltonian describing the system can be presented in form

$$\begin{aligned} H &= H_0 + H_{\rm int} + H_s + H_{sq} + \\ &\quad + H_{T\,0} + H_{\gamma_0} + H_{T\,e} + H_{\gamma_e} + H_{T,\,m} + H_{\gamma_m}, \end{aligned} \tag{2.2a}$$

$$\begin{aligned} H_0 &= \hbar\omega_+ \hat{c}_+^\dagger \hat{c}_+ + \hbar\omega_0 \hat{c}_0^\dagger \hat{c}_0 + \\ &\quad + \hbar\omega_- \hat{c}_-^\dagger \hat{c}_- + \hbar\omega_m \hat{d}^\dagger \hat{d}, \end{aligned} \tag{2.2b}$$

$$\begin{aligned} H_{\rm int} &= \frac{\hbar}{i}\left(\eta\left[\hat{c}_0^\dagger \hat{c}_- + \hat{c}_+^\dagger \hat{c}_0\right]\hat{d} - \right. \\ &\quad \left. - \eta^*\left[\hat{c}_0 \hat{c}_-^\dagger + \hat{c}_+ \hat{c}_0^\dagger\right]\hat{d}^\dagger\right), \end{aligned} \tag{2.2c}$$

$$\hat{H}_s = -F_s x_0 \left(\hat{d} + \hat{d}^\dagger\right), \tag{2.2d}$$

$$H_{sq} = \frac{\hbar\nu}{i}\left(C_{02}\hat{c}_+^\dagger \hat{c}_-^\dagger - C_{02}^* \hat{c}_+ \hat{c}_-\right). \tag{2.2e}$$

Here $\hbar$ is Plank constant. H_0 describes energies of optical modes and mechanical oscillator, $\hat{c}_0$, $\hat{c}_0^\dagger$, $\hat{c}_\pm$, $\hat{c}_\pm^\dagger$ are annihilation and creation operators of the corresponding optical modes, $\hat{d}$, $\hat{d}^\dagger$ are annihilation and creation operators of the mechanical oscillator. The operator of coordinate $\hat{x}$ of the mechanical oscillator is presented in usual form

$$\hat{x} = x_0\left(\hat{d} + \hat{d}^\dagger\right), \quad x_0 = \sqrt{\frac{\hbar}{2m\omega_m}}, \tag{2.3}$$

where m is mass of oscillator. $H_{\rm int}$ is the Hamiltonian of the interaction between optical and mechanical modes[1],

[1] Initially $H_{\rm int} \sim (E_0 + E_+ + E_-)^2 x$, where E_0, E_-, E_+ are electric fields of modes $0, -, +$ on surface of mirror (mass of oscillator), it can be transformed into form (2.2c) after omitting fast oscillating terms.

$\eta \simeq x_0\omega_0/L$ is coupling constant, L is the length of cavity. H_s is a part of Hamiltonian describing signal force F_s. Hamiltonian H_{sq} describes two photon (nondegenerate) internal squeezing, ν is a constant proportional to nonlinear $\chi^{(2)}$ susceptibility of crystal inside cavity, C_{02} is an expectation value of amplitude of classical pump, acting on frequency $2\omega_0$. $H_{T\,0}$ is the Hamiltonian describing the outer environment (regular and fluctuational fields incident on input mirror) and H_{γ_0} is the Hamiltonian of the coupling between the outer environment and the optical modes resulting in decay rate γ_0; $H_{T\,e}$ and H_{γ_e} describe optical losses. The pump is also included into H_{γ_0}. Similarly, $H_{T,\,m}$ is the Hamiltonian of the environment and H_{γ_m} is the Hamiltonian describing coupling between the environment and the mechanical oscillator resulting in decay rate γ_m. See Appendix A for details.

III. ANALYSIS FOR TWO PHOTON NONDEGENERATE INTERNAL AND EXTERNAL SQUEEZING

We denote the normalized input and output optical amplitudes as $\hat{a}_{\pm,\,0}$ and $\hat{b}_{\pm,\,0}$ correspondingly. Using the Hamiltonian (2.2) we derive the equations of motion for the intracavity slow amplitudes of fields (see Appendix A for the full derivation).

$$\dot{\hat{c}}_0 + \gamma\hat{c}_0 = \eta^*\hat{c}_+\hat{d}^\dagger - \eta\hat{c}_-\hat{d} + \sqrt{2\gamma_0}\,\hat{a}_0 + \sqrt{2\gamma_e}\,\hat{e}_0, \quad \gamma = \gamma_0 + \gamma_e \tag{3.1a}$$

$$\dot{\hat{c}}_+ + \gamma\hat{c}_+ = -\eta\hat{c}_0\hat{d} + \nu C_{02}\hat{c}_-^\dagger + \sqrt{2\gamma_0}\,\hat{a}_+ + \sqrt{2\gamma_e}\hat{e}_+, \tag{3.1b}$$

$$\dot{\hat{c}}_- + \gamma\hat{c}_- = \eta^*\hat{c}_0\hat{d}^\dagger + \nu C_{02}\hat{c}_+^\dagger + \sqrt{2\gamma_0}\,\hat{a}_- + \sqrt{2\gamma_e}\,\hat{e}_-, \tag{3.1c}$$

$$\dot{\hat{d}} + \gamma_m\hat{d} = \eta^*(\hat{c}_0\hat{c}_-^\dagger + \hat{c}_0^\dagger\hat{c}_+) + \sqrt{2\gamma_m}\,\hat{q} + f_s. \tag{3.1d}$$

Here $\hat{e}_\pm$ describe quantum fluctuation on account of optical losses, Here $\hat{q}$ is normalized fluctuation force acting on mechanical oscillator, and f_s is normalized signal force (see definition (3.11) below).

The operators $\hat{a}_\pm$, $\hat{e}_\pm$, $\hat{q}$ are characterized with the following commutators and correlators

$$\left[\hat{a}_\pm(t), \hat{a}_\pm^\dagger(t')\right] = \delta(t-t'), \tag{3.2a}$$

$$\left\langle \hat{a}_\pm(t)\,\hat{a}_\pm^\dagger(t')\right\rangle = \delta(t-t'), \tag{3.2b}$$

$$\left[\hat{e}_\pm(t), \hat{e}_\pm^\dagger(t')\right] = \delta(t-t'), \tag{3.2c}$$

$$\left\langle \hat{e}_\pm(t)\,\hat{e}_\pm^\dagger(t')\right\rangle = \delta(t-t'), \tag{3.2d}$$

$$\left[\hat{q}(t), \hat{q}^\dagger(t')\right] = \delta(t-t'), \tag{3.2e}$$

$$\left\langle \hat{q}(t)\hat{q}^\dagger(t')\right\rangle = (2n_T+1)\,\delta(t-t'), \tag{3.2f}$$

$$n_T = \frac{1}{e^{\hbar\omega_m/\kappa_B T} - 1}. \tag{3.2g}$$

Here $\langle\ldots\rangle$ stands for ensemble averaging. Correlators (3.2d), (3.2f) are true for since the fluctuation fields are considered to be in the vacuum (or thermal for $\hat{q}$) state. Correlators (3.2b) is true only while incident field is in coherent state, however it should be revised when incident field is in squeezed state. n_T is thermal number of mechanical quanta, κ_B is Boltzmann constant, T is the ambient temperature.

The relation (3.2b) is given for non-squeezed input amplitudes. For *external* squeezing it should be revised.

The input-output relations connecting the incident ($\hat{a}_\pm$) and intracavity ($\hat{c}_\pm$) amplitudes with output ($\hat{b}_\pm$) amplitudes are

$$\hat{b}_\pm = -\hat{a}_\pm + \sqrt{2\gamma_0}\hat{c}_\pm. \tag{3.3}$$

It is convenient to separate the expectation values of the wave amplitudes at frequency ω_0 (described by block letters) as well as its fluctuation part (described by small letters) and assume that the fluctuations are small:

$$\hat{c}_0 \Rightarrow C_0 + c_0, \tag{3.4}$$

here C_0 stands for the expectation value of the field amplitude in the mode with eigenfrequency ω_0 and c_0 represent the quantum fluctuations of the field in the mode, $|C_0|^2 \gg \langle c_0^\dagger c_0\rangle$. Similar expressions can be written for the optical modes with eigenfrequencies $\omega_\pm$ and the mechanical mode with eigenfrequency ω_m. The normalization of the amplitudes is selected so that $\hbar\omega_0|A_0|^2$ describes the optical power of incident wave [29].

We assume in what follows that the expectation amplitudes are real, same as the coupling constant η, and introduce parametric gain κ

$$C_0 = C_0^*, \quad A_0 = A_0^*, \quad \eta = \eta^*, \tag{3.5a}$$

$$C_{02} = C_{02}^*, \quad \kappa = \nu C_{02} \tag{3.5b}$$

Substituting (3.4) and (3.5) into the equations of motion (3.1) and keeping only terms of first order of smallness we obtain

$$\dot{\hat{c}}_0 + \gamma\hat{c}_0 = \sqrt{2\gamma_0}\hat{a}_0 + \sqrt{2\gamma_e}e_0, \tag{3.6a}$$

$$\dot{\hat{c}}_+ + \gamma\hat{c}_+ = -\eta C_0\hat{d} + \kappa\hat{c}_-^\dagger + \sqrt{2\gamma_0}\,\hat{a}_+ + \sqrt{2\gamma_e}\hat{e}_+, \tag{3.6b}$$

$$\dot{\hat{c}}_- + \gamma\hat{c}_- = \eta C_0\hat{d}^\dagger + \kappa\hat{c}_+^\dagger + \sqrt{2\gamma_0}\,\hat{a}_- + \sqrt{2\gamma_e}\,\hat{e}_-, \tag{3.6c}$$

$$\dot{\hat{d}} + \gamma_m\hat{d} = \eta C_0(\hat{c}_-^\dagger + \hat{c}_+) + \sqrt{2\gamma_m}\,\hat{q} + f_s. \tag{3.6d}$$

Outputs $b_\pm$ around frequencies $\omega_\pm$ have to be detected separately, as shown in Fig. 3. We also see that fluctuation waves around ω_0 do not influence on field components in the vicinity of frequencies $\omega_\pm$ and the first equation (3.6a) separates from the other three, so it is omitted in the further consideration.